AstraZeneca

UNIVERSITY OF MALAYA

ASTRAZENECA CASH FLOW INSIGHTS & FORECASTING



DATATHON 2025 ⟶

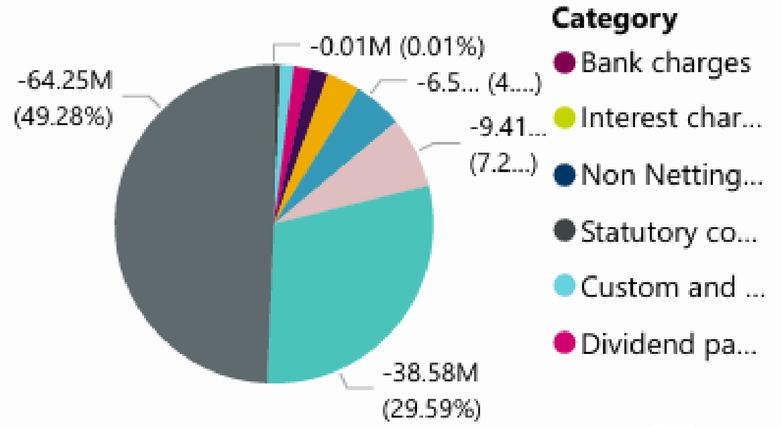
Team Members : Giridarkhanna, Lochanna, Shidasia (Data Mavericks)

01 KPI OF ASTRAZENECA'S CASH FLOW

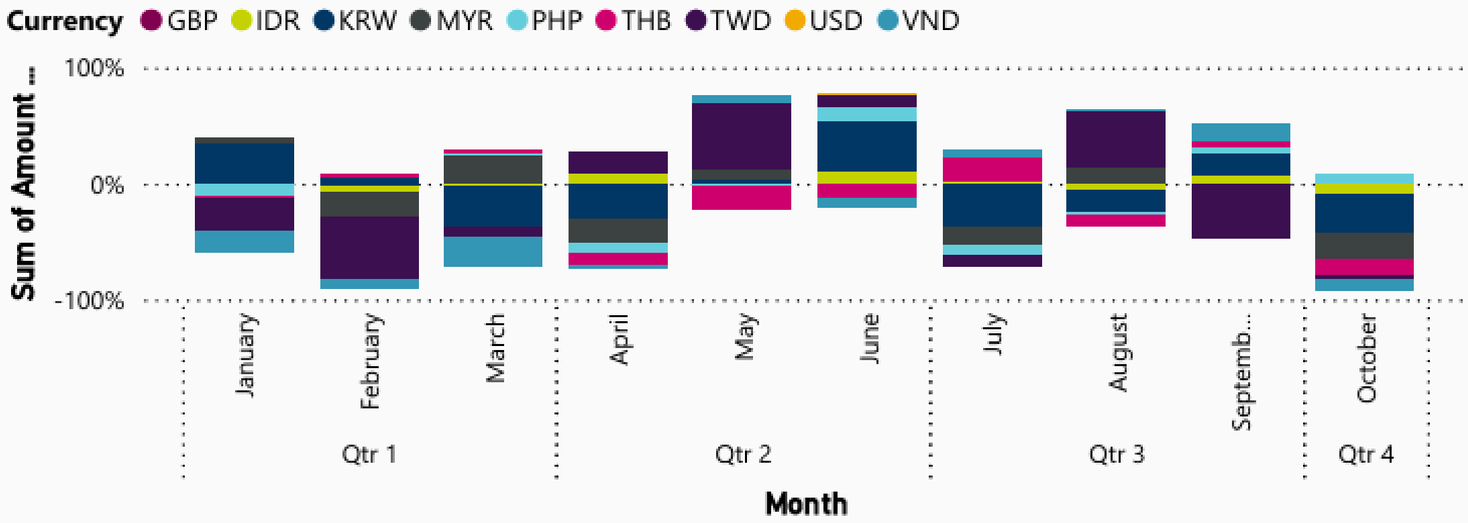
Monitoring Inflows, Outflows, and Overall Liquidity

-3.92M	127.27M	-131.18M
Sum of Amount in USD	Total Inflow	Total Outflow

Total Cash Flow by Category



Total Cash Flow by Month



02 TIME SERIES ANALYSIS OF CASH FLOW

2.1 Actual vs forecast (1 month) for AP, AR, and Tax Payable – short-term cash trends

Category	Total_USD	Frequency
AR	107186649.3	2280
Loan receipt	9447383.157	51
Other receipt	5044653.354	968
Netting AR	4489822.689	51
Non Netting AR	278273.1937	38
Interest income	16429.42928	15
Other	785.28903	12
Bank charges	-8525.67074	8395
Interest charges	-37706.90986	56
Non Netting AP	-93796.54787	15
Statutory contribution	-721947.6649	108
Custom and Duty	-1787612.507	491
Dividend payout	-2215984.181	2
Loan payment and interes	-2353557.549	10
Loan payment	-4420212.321	19
Tax payable	-6511130.673	1168
Payroll	-9407360.491	280
AP	-38575364.43	70456
Netting AP	-64248236.95	113

R Software

Selected Categories (for both 1-month and 6-month forecasts)

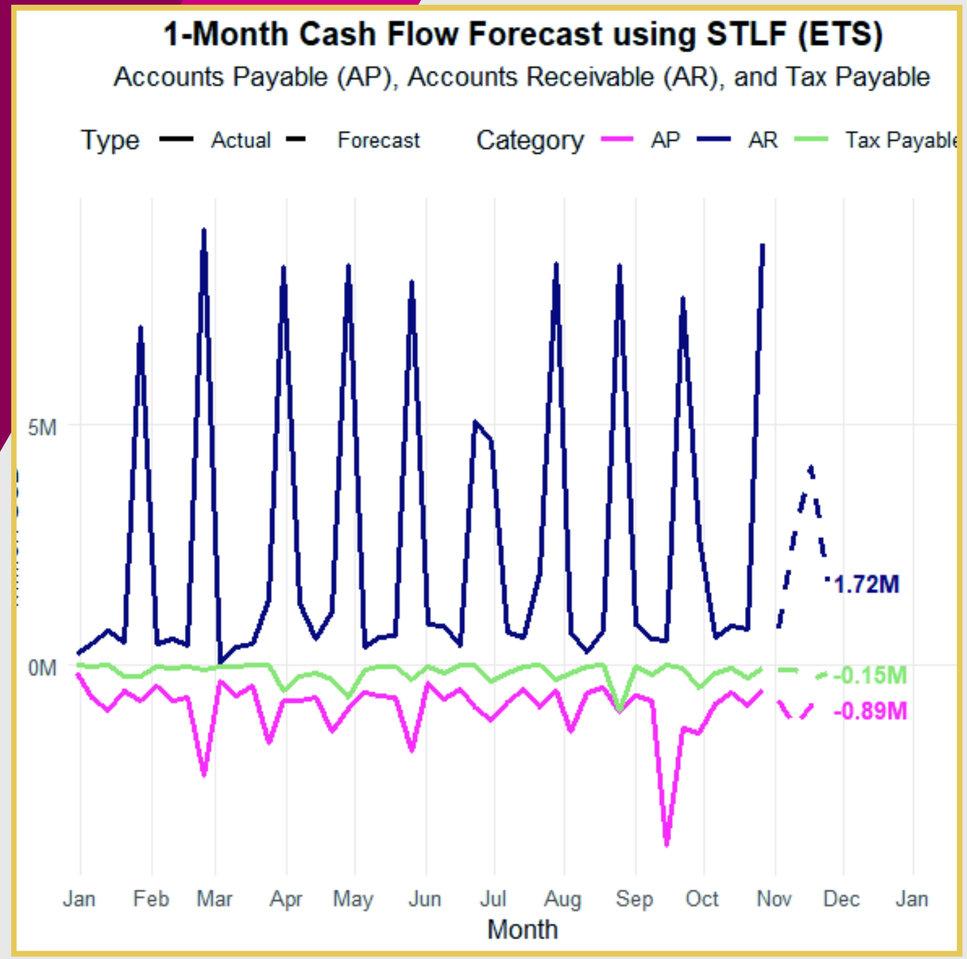
- Accounts Receivable (AR): Highest inflows → drives liquidity
- Accounts Payable (AP): Largest outflows → main cash out risk
- Tax Payable: Predictable statutory outflow → minor but stable

Excluded Categories

- Bank Charges: Minimal impact on net cash

02 TIME SERIES ANALYSIS OF CASH FLOW

2.1 Actual vs forecast (1 month) for AP, AR, and Tax Payable – short-term cash trends




```r
library(forecast); library(ggplot2); library(dplyr); library(lubridate); library(readxl); library(scales)
# Import & prepare data
data <- read_excel("C:/shidasia/.../Datathon Dataset (2).xlsx")
data$Doc..Date <- as.Date(data$Doc..Date)
categories <- c("AP", "AR", "Tax Payable")
weekly_data <- data %>%
  filter(Category %in% categories) %>%
  mutate(Week = floor_date(Doc..Date, "week", week_start = 1)) %>%
  group_by(Week, Category) %>%
  summarise(Total_USD = sum(`Amount in USD`, na.rm = TRUE), .groups = "drop") %>%
  mutate(Type = "Actual")
# Forecast 1-month ahead
forecast_one_month <- function(cat) {
  ts_data <- ts(weekly_data %>% filter(Category == cat) %>% pull(Total_USD), frequency = 4)
  fit <- stlf(ts_data, h = 4, method = 'ets')
  data.frame(
    Week = seq(max(weekly_data$Week) + 7, by = 7, length.out = 4),
    Total_USD = as.numeric(fit$mean),
    Category = cat, Type = "Forecast"
  )
}
forecast_df <- bind_rows(forecast_one_month("AP"), forecast_one_month("AR"), forecast_one_month("Tax Payable"))
plot_data <- bind_rows(weekly_data, forecast_df)

# Plot
ggplot(plot_data, aes(x = Week, y = Total_USD / 1e6, color = Category)) +
  geom_line(aes(linetype = Type), linewidth = 1.2) +
  geom_text(
    data = forecast_df %>% group_by(Category) %>% slice_max(Week, n = 1),
    aes(label = paste0(round(Total_USD / 1e6, 2), "M")),
    hjust = -0.1, vjust = 0.5, size = 3.5, fontface = "bold", show.legend = FALSE
  ) +
  scale_linetype_manual(values = c("Actual" = "solid", "Forecast" = "dashed")) +
  scale_color_manual(values = c("AP" = "magenta", "AR" = "navyblue", "Tax Payable" = "gold")) +
  scale_x_date(date_breaks = "1 month", date_labels = "%b", expand = expansion(mult = c(0.01, 0.2))) +
  labs(title = "1-Month Cash Flow Forecast using STLF (ETS)",
       subtitle = "AP, AR, and Tax Payable", x = "Month", y = "Million USD") +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", hjust = 0.5),
        plot.subtitle = element_text(hjust = 0.5),
        legend.position = "top")
```

Using R software

Model & Assumption

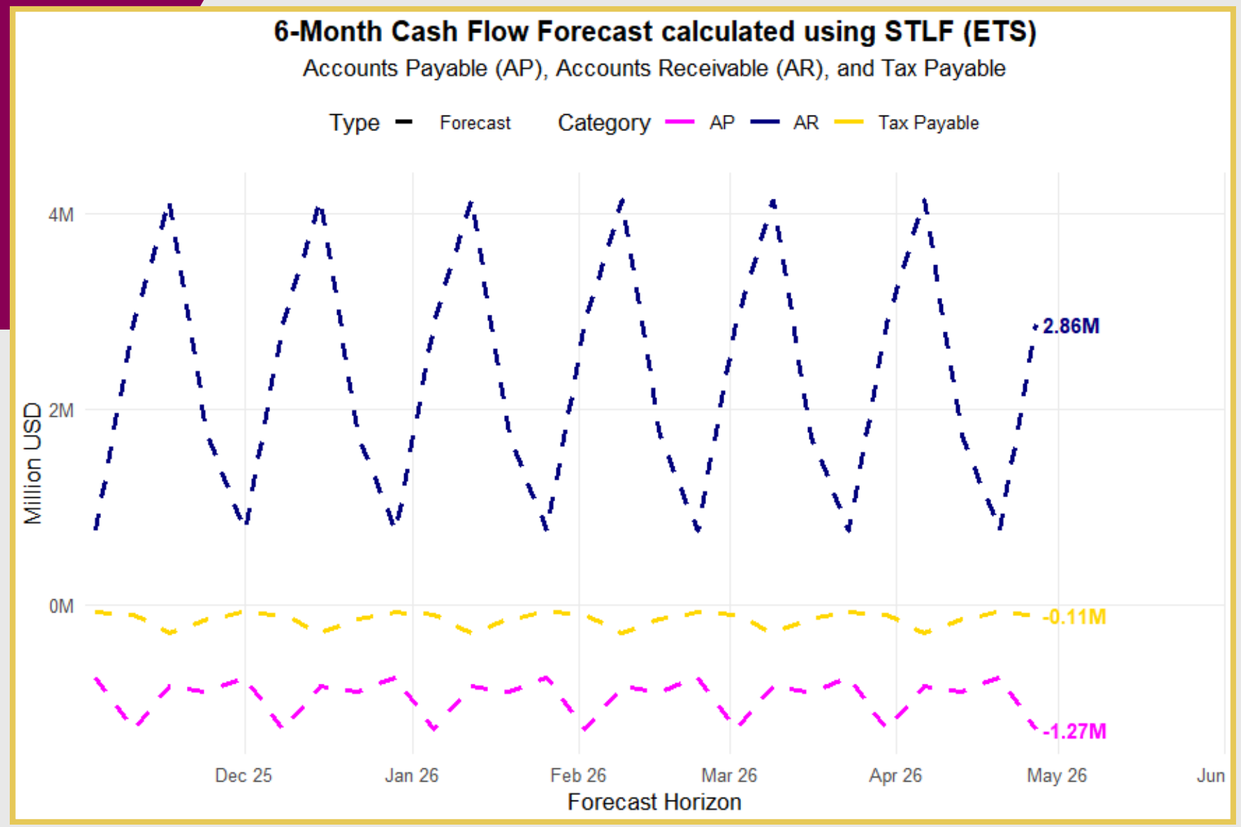
STLF + ETS, assumes historical weekly patterns continue

- Key Drivers Behind Trend :
  - AR: Weekly inflow spikes → boosts liquidity
  - AP: Steady larger outflows → main short-term risk
  - Tax Payable: Stable, predictable outflow → minor impact

- Expected Net Cash Position : Slightly positive; inflows (AR) > outflows (AP + Tax Payable)

02 TIME SERIES ANALYSIS OF CASH FLOW

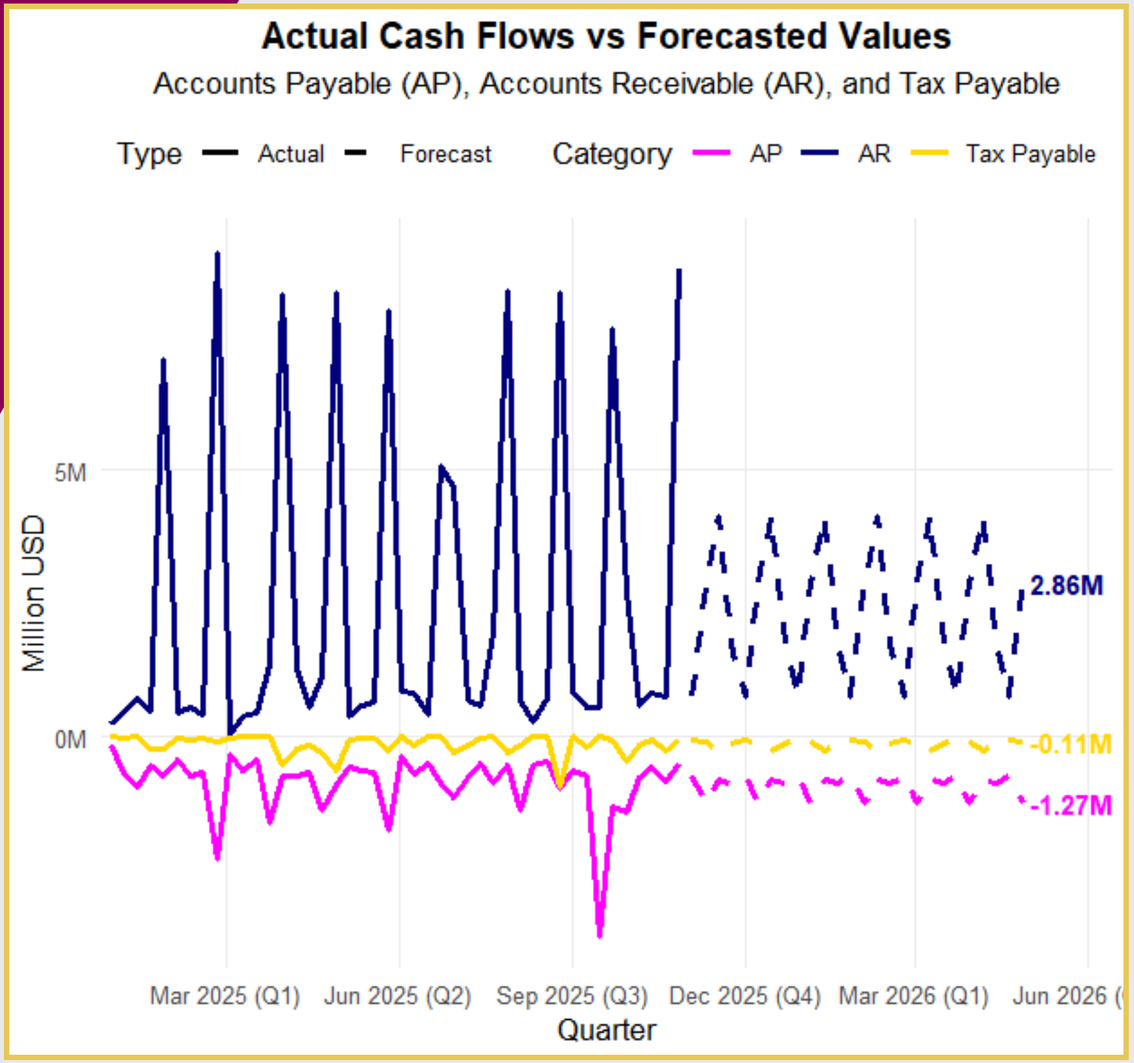
2.2 6-month forecast for AP, AR, and Tax Payable – Long-term cash outlook



- 26-week weekly forecast using STLF (ETS)
- AR: Strong, regular weekly inflows
- AP: Consistent operational outflows
- Tax Payable: Small and stable
- AR inflows > AP + Tax Payable → sustainable medium-term cash position
- Liquidity depends on timely AR collections
- AP is the main structural cash outflow to monitor

02 TIME SERIES ANALYSIS OF CASH FLOW

2.3 Actual vs 6-month forecast (quarterly) – comparing trends and forecast accuracy



Forecast Performance by Category

- Accounts Receivable (AR): Forecast follows weekly spikes; occasional underestimation during high inflow periods
- Accounts Payable (AP): Forecast closely matches outflow levels with minor short-term deviation.
- Tax Payable: Highest accuracy due to stable and predictable pattern

Comparison of forecast vs actual

- Forecast closely follows direction and seasonality of actual cash flows.
- Deviations mainly occur during high-volatility AR inflow periods.

Forecast accuracy evaluation

- Accuracy assessed using MAE, RMSE, and MAPE.
- Highest accuracy: Tax Payable
- Moderate accuracy: AP
- Lowest accuracy: AR (due to inflow volatility)

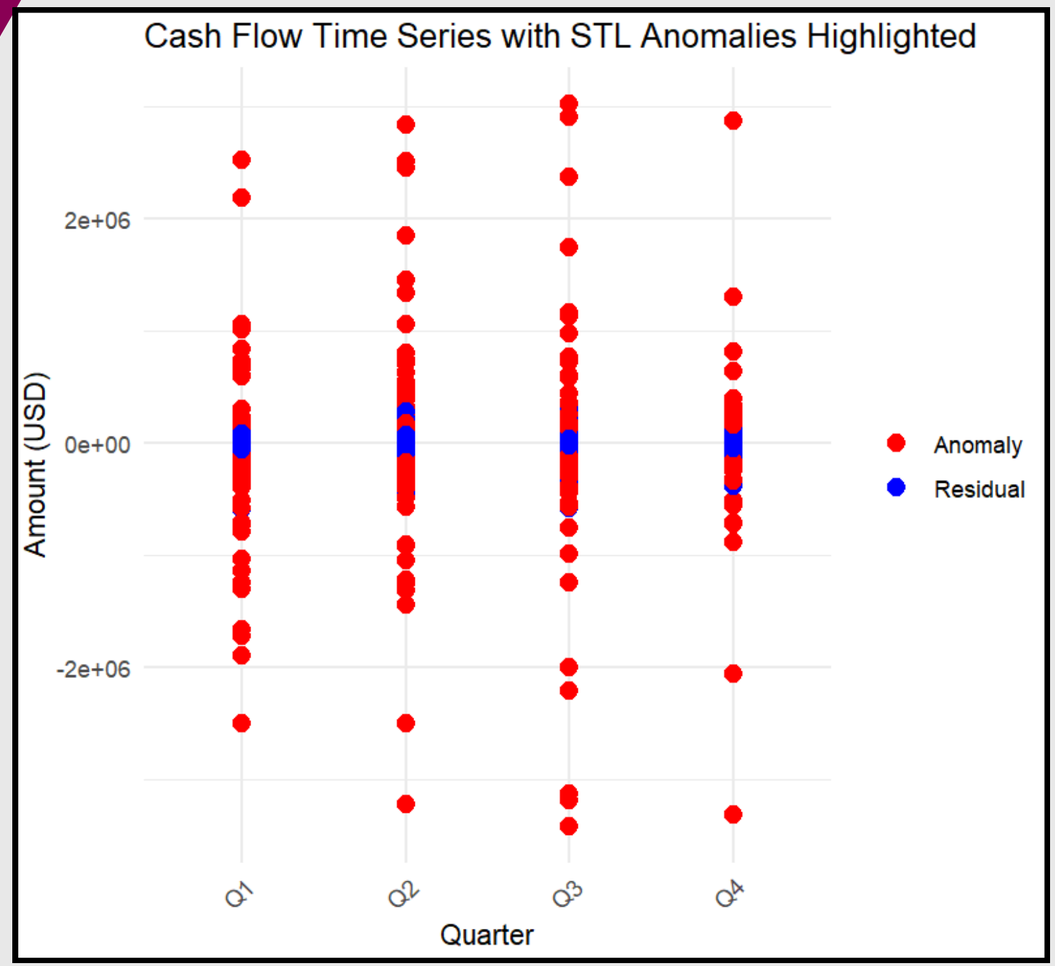
Forecast performance & limitations

- Suitable for liquidity planning, not exact weekly prediction.
- Assumes stable historical patterns; unexpected events not fully captured.

Cash inflow & outflow dependency

- Company liquidity is highly dependent on AR inflows.
- AP is the largest and most consistent cash outflow.
- Tax Payable is fixed and reduces flexibility.

03 ANOMALY DETECTION




```r
# Load necessary packages
library readxl)
library forecast)
library ggplot2)  # To handle date and extract quarter informattion

# Load the dataset (replace with your actual file path)
Mavericks_Copy_of_Datathon_Dataset <- read_excel("UMDAC CATATHON/Mavericks Copy of Datathon Dataset.xlsx")
View(Mavericks_Copy_of_Datathon_Dataset)

# Convert 'Pstng Date' to Date type (adjust format if necessary)
Mavericks_Copy_of_Datathon_Dataset$"Pstng Date" <- as.Date(Maverks_Copy_of_Datathon_Dataset$ "Pstng Date", format
="YY-m-0d")
# Filter relevant columns (Pstng Date and Amount in USD)
data_filtered <- Mavericks_Copy_of_Datathon_Dataset[, c("Psting Date", "Amount in USD"])
head(data_filtered)

# Check the first few rows of the feted data
data_filtered$Quarter <- paste0("Q", quarter(datifatefiltered$'Pstng Date"))
head(data_filtered)

# Anomaly Detection using STL (Seasonal and Trend decomposition)
ts_data <- ts(data_filtered$"Amount in USD", frequency = 52, start = (2025, 1))

# Perform STL decompsition (Seasonal and Trend decomposition using Loess)
decomposition <- stl(ts_data, s.window="periodic")
residuals <- decomposition$time.series[, tinc=

# Add dashed threshold lines to show anomaly cut-off
geom_hline(yintercept = ((-threshold_stl, threushol_stl), l line linetype = 'dashed', color = "blue')

# Plot the original time series with anomalles highlighted in red (based on residuals)
ggplot(data_filtered, aes(x = Quarter, y = "Amount in USD, color = Anomaly)) +- geom_point(size = 3) +
ggtile('Residuals (Seasonality + Trend Removed)','Anomaly', 'Residual')
```

Using R software

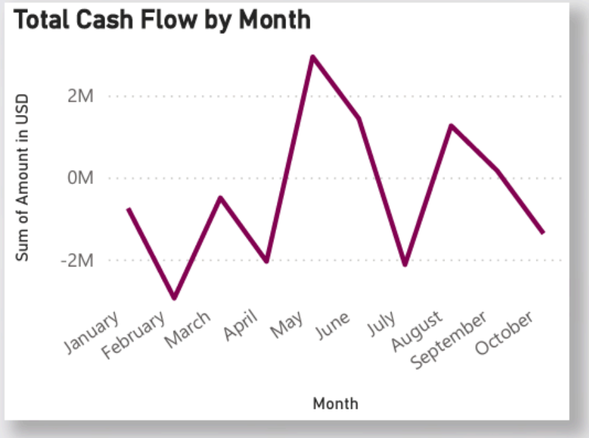
Interesting Findings :

- Unusually high spikes in Q2
- Sudden drop in Q3
- Identify Patterns for Future Projections

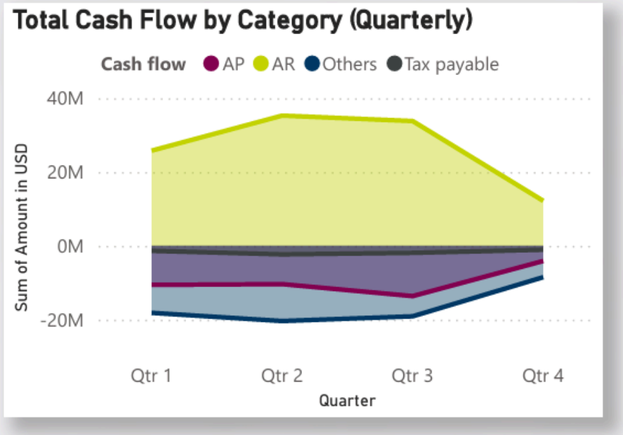
Total Cash Flow by Countries



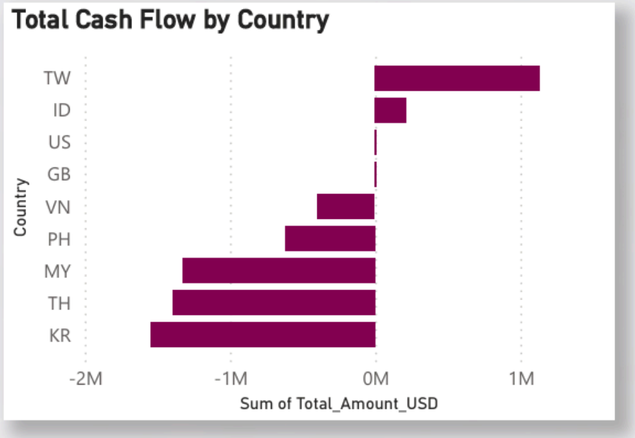
Total Cash Flow by Month



Total Cash Flow by Category (Quarterly)



Total Cash Flow by Country



04 CONCLUSION

1. **Weekly Cash Flow Monitoring**
- monitoring cash flow on weekly basis.
- quick identification in trends or issues.
2. **Quick Identification of Irregularities**
- inclusion of actual vs forecast helps the team to address irregularities faster.
- usage of Z-score as threshold ensures quick corrective actions.
3. **Intuitive Visual Insights**
- easier for analysis to identify high risk categories.
- geographical map allows easy management of international. operations and currency fluctuations.
4. **Proactive Financial Management**
- adjusting strategies based on trends.
- improving liquidity management.



THANK YOU

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